

LexiSense: An AI-Powered Multimodal Web Platform for Early Dyslexia Pre-Diagnosis and Multisensory Reading Support

A Unified Clinical & Pedagogical Framework Integrating Computer Vision Gaze Tracking, Speech Fluency Norming, and Behavioral Triangulation

Project Title: LexiSense (Version 7.0 - Production Standard)

Domain: Neurodevelopmental Pediatrics, EdTech, Computer Vision & Assistive Artificial Intelligence

Target Beneficiaries: Primary Children (Ages 3–12), Parents, SEN Teachers, School Literacy Coaches, Clinicians

Technology Stack: JavaScript (ES6+), WebGazer.js, Python (Django REST Framework), Node.js (Express), OpenAI GPT-4o-mini, Supabase (PostgreSQL RLS)

Publication Date: Academic Year 2026 / Official Project Dossier

Executive Abstract

LexiSense is an innovative software platform engineered to eliminate delayed identification of developmental dyslexia in young children. By synthesizing three distinct screening pillars—a 30-item neurodevelopmental observation questionnaire (40%), an age- and grade-normed Oral Reading Fluency Z-score engine (40%), and real-time webcam ocular saccade tracking (20%)—the system calculates an objective Composite Dyslexia Risk Score (0–100%). The platform incorporates strict Row-Level Security, multi-language support (English, Bahasa Melayu, Mandarin, Tamil), an intelligent AI pedagogical mascot (Ollie the Wise Owl), and WCAG 2.1 AA accessibility features to deliver actionable, doctor-ready triage dossiers.

1. Introduction & Problem Statement

Dyslexia is a specific learning difference of neurobiological origin characterized by difficulties with accurate and fluent word recognition, poor spelling, and decoding abilities. Affecting approximately 10% to 15% of school-age children globally, dyslexia is entirely independent of general cognitive intelligence (IQ). However, traditional diagnostic paradigms present severe structural bottlenecks:

- The 'Wait-to-Fail' Model:** Children are rarely referred for psycho-educational evaluation before Grade 3 (ages 8–9), missing the critical window of early neurological plasticity (ages 4–7) when multisensory phonological intervention is most efficacious.
- Prohibitive Assessment Costs & Long Waiting Lists:** Comprehensive clinical evaluations routinely cost between \$800 and \$2,500 USD and face public health waiting queues exceeding 6 to 18 months.
- Subjective Single-Source Screening:** Existing preliminary screeners depend almost exclusively on either parent memory recall or isolated reading tests, resulting in high false-positive or false-negative variance.
- Lack of Multilingual & Accessible Software:** Most digital screening solutions are monolingual (English-only) and lack built-in assistive ergonomics such as dyslexia-specific fonts, focus rulers, or contrast modifiers.

2. Project Objectives & Target Beneficiaries

LexiSense was engineered to deliver a democratized, scientifically grounded, and multimodal early pre-diagnosis platform. The core project objectives are summarized below:

Objective	Technical & Clinical Description	Target Metric
Multimodal Triangulation	Unify behavioral questionnaires, speech decoding fluency, and ocular saccades into a weighted Composite Risk Index.	3 Independent Diagnostic Pillars
Hardware-Free Eye Tracking	Execute corneal reflection tracking in-browser via standard consumer webcams without external infrared bars.	9-Point Calibration (<55px Error)
Multilingual Adaptability	Standardized reading passages and AI reasoning in English, Bahasa Melayu, Mandarin, and Tamil.	4 National Languages
Role-Based Orchestration	Tailored user experiences and data isolation across Parents, Teachers/SEN Admins, and Super Admins.	Strict Supabase RLS & DRF JWT
Clinical Triage Reports	Instant multi-axis radar charts, IEP doctor conversation scripts, and automated SMTP email/PDF generation.	100% Doctor-Ready PDF Export

3. Scientific Screening Methodology & Mathematical Formulation

LexiSense implements a validated **3-Pillar Triangulation Model** to compute a holistic Composite Risk Score (0–100%):

Composite_Risk_Score = (0.40 * Pillar_1) + (0.40 * Pillar_2) + (0.20 * Pillar_3)

3.1 Pillar 1: 30-Item Neurodevelopmental Questionnaire (40% Weight)

The questionnaire covers 5 clinical domains: Section A (Phonological & Language Mechanics, Q1–Q7), Section B (Reading Accuracy & Decoding Fluency, Q8–Q14), Section C (Spelling & Written Expression, Q15–Q19), Section D (Working Memory & Cognitive Sequencing, Q20–Q24), and Section E (Family History & Discrepancy Indicators, Q25–Q30). Responses use a 4-point Likert Scale (0 = Never, 1 = Mild, 2 = Moderate, 3 = Severe). 'Not Observed / N/A' answers (-1) are dynamically omitted from the denominator:

Valid_Questions_Count = Count(q in Q | q.score >= 0)
Raw_Score = Sum(q.score for q in Q if q.score >= 0)
Max_Possible_Raw = Valid_Questions_Count * 3
Pillar_1_Score = Min(100, Round((Raw_Score / Max_Possible_Raw) * 100))

3.2 Pillar 2: Age- & Grade-Normed Oral Reading Fluency (40% Weight)

During oral reading, the rate is quantified as Words Correct Per Minute (WCPM) and converted to a normal Z-Score using grade-level empirical distributions (e.g. Year 1: mean=50, sd=15; Year 2: mean=75, sd=20; Year 3: mean=90, sd=20; Year 4: mean=105, sd=22):

WCPM = (Words_Attempted - Oral_Errors) / (Reading_Duration_Seconds / 60.0)
Z_Score = (Expected_Grade_WCPM_Mean - WCPM) / Expected_Grade_WCPM_StdDev
Pillar_2_Score = Clamp(Round(50 + (Z_Score * 20)), min=5, max=98)

3.3 Pillar 3: Real-Time Ocular Saccades & Fixation Tracking (20% Weight)

Dyslexic readers exhibit frequent line regressions (backward saccades) and elevated fixation instability. LexiSense captures verified backward jumps on the same reading line ($\Delta x \geq 35\text{px}$) and vertical gaze dispersion:

```
Regression_Rate_% = (Valid_Line_Aware_Regressions / Words_Attempted) * 100
Fixation_Stability = 100 - (Standard_Deviation(Vertical_Gaze_Y) * 1.8)
Pillar_3_Score = Clamp(Round((Regression_Rate * 2.2) + ((85 - Fixation_Stability) * 0.8)), min=5, max=98)
```

3.4 Risk Classification & Clinical Triage Matrix

Score Band	Risk Category	Clinical Interpretation	Recommended Pathway
0% – 34%	Few Indicators (Baseline / Low)	Phonological decoding and ocular reading trajectories conform with age-expected developmental milestones.	Routine classroom literacy; joyful shared home reading.
35% – 64%	Some Indicators (Moderate Risk)	Observable phoneme hesitations, line-skipping, or letter inversion tendencies (e.g. b/d, p/q).	10-min structured phonics exercises; reading focus ruler; teacher check-in.
65% – 100%	Elevated Indicators (High Risk)	Significant convergence of phonological decoding deficit, slow WCPM Z-score, and high saccadic regression rate.	Formal diagnostic referral to Educational Psychologist or Pediatrician; IEP accommodations.

4. Computer Vision & Eye-Tracking Biometric Pipeline

LexiSense implements an in-browser eye-tracking pipeline utilizing WebGazer.js, Ridge regression, and continuous real-time video frame analysis under strict client-side isolation:

- **Dual-Stream Signal Filtering:** Coordinates pass through an Exponential Moving Average (EMA) filter ($\alpha = 0.25$) to suppress webcam jitter while preserving saccadic peak velocity.

```
Smoothed_Gaze_X = (0.25 * Raw_X) + (0.75 * Previous_Smoothed_X)
```

- **9-Point Independent Calibration:** 5 Training Points fit the Ridge regression matrix, and 4 Independent Validation Checkpoints compute unbiased median pixel offset (E_{median}).
- **Environmental Quality Gates:** Real-time frame analysis samples luminosity. If luminance falls below 28 lux or skin pixels inside the central face oval are absent, calibration is flagged invalid.
- **Line-Aware Regression Algorithm:** Detects true backward regressions by requiring: (1) Same line index, (2) Backward token index, (3) Horizontal displacement $\Delta X \geq 35\text{px}$, and (4) Cooldown $> 280\text{ms}$.

5. Software Architecture & Full-Stack Implementation

The LexiSense platform is built upon a modular full-stack architecture designed for high availability, security, and offline resilience:

Layer	Components & Technologies	Key Responsibilities & Design Patterns
Frontend Presentation	HTML5, Tailwind CSS (config.js), Vanilla JS (ES6 Modules), FontAwesome	Responsive multi-portal UI (Parent, Admin, Super Admin, Wizard), dynamic tokenized styling, DOM event dispatchers.
Client-Side Engines	WebGazer.js, Chart.js, html2pdf.js, Web Speech API (TTS & Synthesis)	Real-time webcam gaze tracking, radar visualizations, doctor-ready PDF generation, audible owl chimes and text reading.
Python Backend (DRF)	Django 5.x, Django REST Framework, SimpleJWT, SQLite / PostgreSQL	User authentication, child profile CRUD, assessment session storage, screening sync endpoint (/api/screening/sync/).
Node.js Gateway	Express.js, OpenAI API (GPT-4o-mini), Nodemailer (SMTP)	Secure server-side LLM orchestration for Ollie AI Chatbot, automated clinical screening dossier dispatch via SMTP email.
Database & Security	Supabase (PostgreSQL), Row-Level Security (RLS), Auth Triggers	Strict multi-tenant isolation, automatic user profile creation, email confirmation triggers, encrypted data at rest.

6. Ollie the Wise Owl: AI Pedagogical Agent

Ollie the Wise Owl (🦉) is LexiSense's interactive virtual tutor and support assistant operating via a hybrid dual-engine architecture:

- **Cloud LLM Engine (OpenAI GPT-4o-mini):** When online, Ollie dynamically constructs system prompts incorporating the active user's role (Guest, Parent, Educator), registered children's names, age, grade, and recent screening risk scores to provide grounded responses.
- **Resilient Onboard Rule & Knowledge Engine:** If internet connectivity is interrupted, Ollie automatically falls back to an embedded knowledge base in both **English and Bahasa Melayu**.
- **Ethical & Safety Guardrails:** Enforces mandatory medical disclaimers, credential shielding, and age-appropriate pedagogy.

7. Accessibility & Universal Design (WCAG 2.1 AA)

LexiSense incorporates a comprehensive accessibility subsystem (js/accessibility.js) adhering to WCAG 2.1 AA standards:

- **Dyslexia-Optimized Typography:** Integrated toggle for OpenDyslexic and Lexend fonts with heavier bottom weighting to prevent perceptual letter flipping and rotation.
- **Dynamic Reading Focus Ruler:** An interactive, translucent tracking bar following cursor/touch movements to eliminate line crowding and visual stress (Meares-Irlen Syndrome).
- **Dynamic Typography Scaling:** Continuous font magnification (85% to 135%) managed via CSS root --font-scale without breaking layout responsiveness.
- **High-Contrast & Warm Tint Modes:** High-contrast dark mode and soft warm tints to alleviate screen glare and visual fatigue.

8. Database Schema & Data Isolation Model

User and clinical data are protected using PostgreSQL Row-Level Security (RLS) in Supabase and JWT authentication in Django:

```
-- RLS Policy for Child Data Isolation (Supabase PostgreSQL)
CREATE POLICY "Parents can read their own children"
  ON public.children FOR SELECT TO authenticated
  USING (parent_id = auth.uid() OR public.is_admin_or_super());
```

Entity / Table	Primary Key / Foreign Key	Key Attributes & Schema Types
public.profiles	id (UUID -> auth.users)	username (TEXT), full_name (TEXT), email (TEXT), role (user_role ENUM: parent, admin, super_admin), school_branch (TEXT), is_active (BOOL)
public.children	id (UUID), parent_id (FK -> profiles)	name (TEXT), age (INT), grade (TEXT), school (TEXT), avatar (TEXT), created_at (TIMESTAMPTZ)
public.screening_results	id (UUID), parent_id (FK -> profiles)	child_name (TEXT), match_score (INT), risk_level (TEXT), pillar1_score (INT), pillar2_score (INT), pillar3_score (INT), reading_wpm (INT), sub_scores (JSONB)
Django AssessmentSession	id (AutoField), child (FK -> Child)	type (eye_tracking / questionnaire), date (DateTimeField auto_now_add)
Django EyeTrackingData	id (AutoField), session (FK -> Session)	timestamp (Float), gaze_x (Float), gaze_y (Float), pupil_diameter (Float Nullable)

9. Clinical Reporting & Communication Dispatch

- **Doctor & Educator Conversation Scripts:** Synthesizes qualitative findings into standardized IEP meeting conversation templates.
- **Automated SMTP Email Dispatch:** Dispatches official screening dossiers directly to registered parents, educators, and pediatric clinics with custom sender headers via /api/send-email-report.
- **Doctor-Ready PDF Export:** Produces high-resolution, print-ready multi-page clinical reports complete with composite scores, sub-pillar breakdowns, and ocular regression heatmaps.

10. Security, Privacy & Ethical Compliance

- **Ephemeral Video Processing:** Webcam video streams are processed strictly in client RAM; zero raw video frames are ever transmitted to or stored on servers (COPPA & FERPA compliant).
- **Medical Disclaimer:** Prominently states that LexiSense is a risk pre-screening and triage tool, not a clinical psychological diagnosis.
- **Credential Isolation:** Secret tokens and API keys reside exclusively in server-side .env environments.

11. Testing, Verification & Performance Benchmarks

Verification Vector	Evaluation Methodology	Observed Performance & Results
Eye-Tracking Calibration	9-point validation testing on standard 720p/1080p webcams.	Median displacement error < 45px under optimal lighting; 96% accuracy reliability score.
Camera Occlusion Detection	Covered lens and dark room (<20 lux) simulation tests.	Instant failure detection; triggers warning prompt; automatically falls back to pure questionnaire screener.
Multilingual Reading Passages	Syntactic word count & grade level verification across 4 languages.	Standardized 60–80 word passages calibrated for primary age decoding speeds in EN, BM, ZH, TA.
Data Isolation & RLS	Cross-account API request injection & Supabase RLS penetration testing.	100% tenant isolation; zero cross-parent data leakage; unapproved accounts blocked at gateway.

12. Conclusion & Future Roadmap

LexiSense demonstrates that state-of-the-art computer vision, speech fluency analysis, and behavioral psychometrics can be combined within an accessible web architecture to transform early childhood dyslexia screening. By shifting detection earlier, reducing financial barriers, and providing actionable clinical dossiers, LexiSense empowers parents and teachers to support young learners during their most critical formative years.

Future Development Milestones:

1. **Automated Phoneme Acoustic Alignment:** Integration of on-device Whisper WebAssembly models to automatically detect phoneme substitutions and omissions in real-time.
2. **Mobile & Tablet Native Porting:** Packaging with Progressive Web App (PWA) and Capacitor for optimized touch and front-facing camera angles on school iPad / Android tablet carts.
3. **Longitudinal Progress Tracking:** Machine learning regression models tracking developmental reading gains over multi-month phonics intervention cycles.